LIFE WITHOUT A SPOROPHYTE: THE ORIGIN AND GENOMIC CONSEQUENCES OF ASEXUAL REPRODUCTION IN A GAMETOPHYTE-ONLY FERN

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Premise of research. The vast majority of eukaryotes reproduce by sexual reproduction, which is accompanied by meiotic recombination. Asexual reproduction, including the loss of meiosis, has been well characterized in several animal lineages but is vastly understudied in plants and particularly in the gametophyte life stage. Here, we explore the genomic consequences of strict clonal reproduction in the gametophyte-only fern *Vittaria appalachiana*.

Methodology. Using publicly available transcriptome assemblies, we assessed how asexual reproduction can alter genomic features by comparing the transcriptomes of the asexually reproducing *V. appalachiana* (Pteridaceae) with sexually reproducing species in the same family, including congener *Vittaria lineata*. We explored several hypotheses relating asexual reproduction to the efficacy of purifying selection, transposable element load, GC-biased gene conversion, and the role of whole-genome duplications (WGDs).

Pivotal results. We found putative support for predicted genomic consequences of asexual reproduction, including decreased efficacy of purifying selection (particularly in genes related to the cell cycle), altered transposable element load, and decreased GC content in *V. appalachiana*. We identified two putative WGDs, shared by Pteridaceae and the most recent common ancestor of the *Vittaria* genus, which may have contributed to *V. appalachiana*'s persistence over evolutionary time without sexual reproduction.

Conclusions. *Vittaria appalachiana* is a gametophyte-only fern that is well suited to address fundamental questions regarding the long-term genomic effects of asexual reproduction in ferns, where this phenomenon has had little scientific attention in plants. This initial exploration into the genomic consequences of asexual reproduction is one of the first in ferns and highlights several avenues for future research.

Keywords: gametophyte, comparative transcriptomics, asexual reproduction, polyploidy, *Vittaria appalachiana*, Pteridaceae, fern.

Online enhancements: supplemental figure, appendixes.

Introduction

The evolution and maintenance of sexual reproduction have long been of great interest for evolutionary biologists; sex is a costly phenomenon (e.g., costs of males [Smith 1978], genome dilution [Williams 1975]; reviewed in Lehtonen et al. 2012), yet it is the predominant reproductive system in eukaryotes (Otto 2008; Tree of Sex Consortium 2014). Many hypotheses regarding the persistence of sex in natural systems have been proposed, including the Red Queen hypothesis (coevolution between hosts and parasites; e.g., Levin 1975; Howard and Lively 1994), group selection (Nunney 1989), and the purging of deleterious alleles (e.g., Felsenstein 1974; reviewed by Hurst and Peck 1996). Perhaps one of the most powerful and important aspects driving the evolution and maintenance of sexual reproduction is the process of meiotic recombination, which shapes genome evolution (e.g., Lynch 2007; Melamed-Bessudo et al. 2016). During meiosis, homologous chromosomes exchange genetic material, which can break up linked loci on the same chromosome. Recombination can be viewed as a force that can simultaneously facilitate adaptation by splitting up combinations of deleterious alleles (or creating combinations of beneficial alleles) and as a detrimental force that disrupts beneficial sets of alleles. Meiotic recombination, however, is absent in clonally reproducing organisms (hereafter, asexually reproducing organisms), and the lack of recombination has been suggested to have several profound theoretical consequences, including

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(1) decreased efficacy of purifying selection, (2) altered transposable element (TE) load, and (3) decreased GC bias. We describe the theory and evidence for each of these consequences below.

Decreased Efficacy of Purifying Selection

Selection on one locus can interfere with selection on other loci by preventing fixation of beneficial mutants; the strength of this interference depends on the degree of linkage between loci under selection (Muller 1964; Hill and Robertson 1966; Felsenstein 1974; see Comeron et al. 2008 for a full review). In sexually reproducing organisms, recombination works at each reproductive cycle to break up linked loci. This process can break apart favorable allelic combinations but also allows for the purging of deleterious alleles. In asexually reproducing organisms, however, all loci across the genome become linked simultaneously when sexual reproduction is lost and new combinations of alleles cannot be generated (with the exception of infrequent mitotic recombination and mutation), making it essentially impossible to purge deleterious alleles individually. The accumulation of deleterious mutations in the genomes of asexually reproducing organisms and subsequent fitness decreases has been termed Muller's ratchet (Muller 1964). In effect, asexually reproducing taxa are unable to overcome the accumulation of deleterious mutations and will eventually succumb to the effects of these alleles and ultimately go extinct. The accumulation of deleterious mutations has been observed in several apomictic and parthenogenically reproducing animals (including stick insects [Bast et al. 2018] and freshwater snails [Johnson and Howard 2007; Neiman et al. 2010]) and in asexual plants (*Oenothera* [Hollister et al. 2015]). However, Brandt et al. (2017) identified a case of effective purifying selection in ancient asexual oribatid mites, suggesting that there may be other factors influencing selection in asexually reproducing taxa (e.g., cryptic sex or nonmeiotic recombination).

Altered TE Load

TEs are sequences that can move around the genome via either a "cut-and-paste" (DNA transposons) or "copy-and-paste" (retrotransposons or retroelements) mechanism (Bourque et al. 2018). The movement of TEs through the genome can cause deleterious effects by inserting into coding or regulatory sequences (Sigman and Slotkin 2016). In sexually reproducing populations, TEs can easily spread and proliferate (Hickey 1982); however, recombination and selection can act to remove them from the population. Without the exchange of genetic material between individuals (i.e., when only asexual reproduction occurs), TEs may not spread as freely; however, without the recombination that accompanies sex, TEs also cannot be purged as efficiently as they can in sexually reproducing organisms (or at all). This suggests that TEs could build up in the genome and eventually lead to extinction in asexual lineages (Arkhipova and Meselson 2005). In contrast, asexually reproducing lineages with few or no active TEs and/or effective mechanisms to suppress TE movement would be able to avoid extinction (Hickey 1982). Experimental tests in yeast (Bast et al. 2016, 2019) and data from natural populations of parthenogenetic animals (Arkhipova and Meselson 2000; Jaron et al. 2021) have found that asexually reproducing taxa generally have lower TE loads than sexually reproducing taxa. Jaron et al. (2021) further suggest, however, that lineage-specific factors may confound the presence of TE loads in parthenogenic animals. For example, a clade of sexually reproducing organisms may have an unusually high level of TEs; any member of that clade that loses sexual reproduction would inherit that high level of TEs from its sexually reproducing ancestor, despite the prediction that TE loads should be low in asexual organisms. The converse may also be true, where a member of a clade that loses sexual reproduction inherits low TE levels as well as the mechanisms to suppress future TE activity.

Decreased GC Content

Crossover between chromosomes during meiosis is initiated by the nonrandom generation of double-stranded breaks throughout the genome. Homologous regions on chromosome pairs can then be swapped between these partners. An unequal crossing-over event where the gene sequence of one partner replaces the sequence of the other partner, without a corresponding swap, is called gene conversion. The mechanisms that repair double-stranded breaks can favor the genetic material of one chromosome partner (i.e., an allele) over another. The preferential incorporation of G and C base pairs during meiotic recombination specifically is called GC-biased gene conversion (gBGC; Marais 2003; Duret and Galtier 2009) and leads to increases in GC content. gBGC is found across eukaryotes (Marais 2003; Pessia et al. 2012) but has been more extensively studied in animals than in plants. Decreases in recombination rate have also been correlated with decreased GC content in humans and chickens (Duret and Galtier 2009). In rice, Muyle et al. (2011) found that selfing taxa had significantly lower GC content than their outcrossing counterparts, which they attributed to less efficient recombination and lower rates of gBGC. Furthermore, some asexually reproducing animals also show this trend of decreased GC content compared to sexually reproducing taxa (e.g., stick insects; Bast et al. 2018).

In addition to the direct potential consequences of the lack of recombination described above, asexual reproduction may further be expected to drive gene family evolution. Gene families involved in sexual reproduction and meiotic processes are expected to undergo contractions through mutational accumulation due to relaxed purifying selection (Schurko and Logsdon 2008; Schwander et al. 2013). The core genes involved in meiosis are conserved across eukaryotes (de Massy 2013) and have functions related to chromosome pairing, double-stranded break repair, and recombination (Schurko and Logsdon 2008). Evidence for losses in these meiosis-related genes in asexually reproducing organisms, however, is mixed, and perhaps surprisingly, there is no clear pattern of gene loss in asexually reproducing animals (reviewed in Jaron et al. 2021). To date, similar patterns have not been thoroughly investigated in plants, although recent work has suggested that asexual reproduction alters gene expression of meiosis-related genes (e.g., in *Brassica rapa*; Braynen et al. 2017) and results in changes to the epigenomic landscape, including differential methylation of TEs (reviewed by Grusz 2016) in plant genomes.

A possible mechanism that can allow asexually reproducing organisms to escape the consequences of shirking sex is polyploidy. Whole-genome duplication (WGD) is a common phenomenon

in plants, with around one-third of speciation events in ferns and 15% in angiosperms associated with changes in ploidy (Wood et al. 2009). Polyploidy is known to be associated with asexual lineages (Otto and Whitton 2000), though a causal relationship is unclear (but see Maciver 2019 for a hypothesis relating the facilitation of asexual reproduction by polyploidy in early eukaryotes). Gene duplications associated with polyploidy could allow for genomic flexibility in asexually reproducing organisms (Blanc-Mathieu et al. 2017). For example, retained copies of genes can mask deleterious alleles, or, following a WGD, one copy of the gene could undergo neofunctionalization in which positive selection alters its function while the other copy retains the original function (Li et al. 2021). Given the widespread occurrence of polyploidy in plants, its co-occurrence with asexual reproduction warrants additional study.

One clade of particular interest for the study of asexual reproduction is ferns. Unlike other vascular plants, ferns (as well as lycophytes) have gametophyte and sporophyte life stages that are nutritionally independent (Haufler et al. 2016). Several fern taxa can persist indefinitely without a sporophyte (e.g., Farrar 1967), a phenomenon termed "separation of generations" (Pinson et al. 2017). The complete loss of the sporophyte phase is accompanied by the loss of sexual reproduction and, therefore, meiotic recombination. Vegetative reproduction in the gametophytic life stage is largely understudied in plants partly because of the ephemeral nature and small size of gametophytes and is therefore a crucial gap in our knowledge of vascular plant evolution. One of the best-known and widespread gametophyte-only ferns is the Appalachian gametophyte, *Vittaria appalachiana* (fig. 1). As its name implies, *V. appalachiana* inhabits limestone

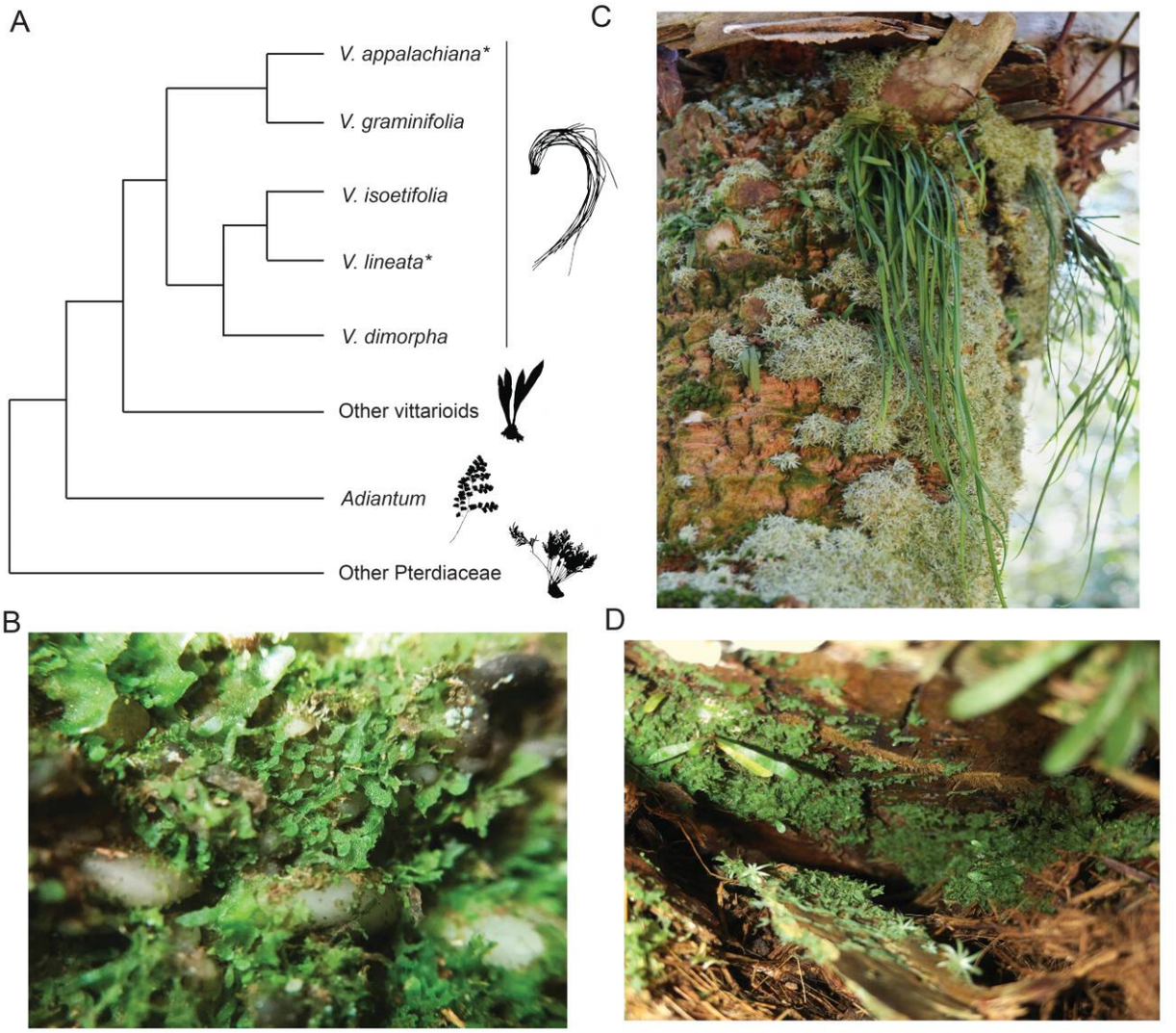


Fig. 1 *A*, Phylogeny of *Vittaria*, Vittarioideae, and Pteridaceae as in Pinson and Schuettpelz (2016) and Testo and Sundue (2016). *Vittaria* highlighted by vertical line delimitation. Asterisks denote focal species in this study. From top, silhouettes of *Vittaria lineata*, *Polytaenium cajenense*, *Adiantum capillus-veneris*, and *Cryptogramma acrostichoides* are available from PhyloPic under the public domain. *B*, Gametophytes of *Vittaria appalachiana* at ×10 magnification. *C*, Sporophytes of *V. lineata*. *D*, Gametophytes of *V. lineata*. All images by J. A. Pelosi.

rockhouses throughout the Appalachian Mountains in eastern North America and exists solely in the gametophyte phase (few aborted sporophytes have been identified in natural and laboratory conditions [Caponetti et al. 1982], but none of these grew taller than a few centimeters or lived long enough to produce spores; reviewed in Pinson et al. 2017). *Vittaria appalachiana* is tetraploid (i.e., the gametophytes are diploid relative to the base number of the genus: n = 2x = 120 [the base number of *Vittaria* is 60]) and is sister to the sexual species *Vittaria graminifolia*, which has both diploid and tetraploid populations (Smith and Mickel 1977; Pinson et al. 2017). The mode of polyploid origin is of open debate, with both autopolyploid (Pinson and Schuettpelz 2016) and allopolyploid (Farrar 1990) possibilities. This asexually reproducing gametophytic species has been the focus of several systematic (Farrar and Mickel 1991; Pinson and Schuettpelz 2016), ecological (Stevens and Emery 2015; Chambers and Emery 2016; Chambers et al. 2017), and population genetic (Farrar 1985, 1990) studies. Despite this framework, many questions remain unanswered regarding the evolution and persistence of this charismatic taxon. Here, we investigate what genomic features (if any) could have been shaped by the loss of sexual reproduction and whether functional changes are associated with asexual genome evolution in the diploid gametophyte-only fern *V. appalachiana* by comparing transcriptomic features to its congener *Vittaria lineata* (also a tetraploid species). We address the role of polyploidy in this lineage and assess the potential effects of asexual reproduction on (1) efficacy of purifying selection, (2) TE load, (3) GC-biased gene conversion, and (4) gene family evolution.

Methods

Transcriptome Sampling

We downloaded transcriptome assemblies from One Thousand Plant Transcriptomes Initiative (1KP; 2019) for 12 fern species in the family Pteridaceae (*Adiantum aleuticum*, *Adiantum raddianum*, *Argyrochosma nivea*, *Cryptogramma acrostichoides*, *Gaga arizonica*, *Myriopteris rufa*, *Notholaena montieliae*, *Pityrogramma trifoliata*, *Pteris ensiformis*, *Pteris vittata*, *Vittaria appalachiana*, and *Vittaria lineata*) and an outgroup taxon (*Asplenium platyneuron*). All transcriptomes were derived from sporophytic tissues except in the case of *V. appalachiana*, which was gametophyte tissue. None of our subsequent analyses involved relative comparisons that would be biased by the use of different life stage transcriptomes. Transcriptomes were passed to TransDecoder version 5.5 (Grabherr et al. 2011) to generate coding sequence (CDS) and proteome files and then annotated with Trinotate version 3.2 (Bryant et al. 2017). Peptide sequences from all transcriptomes were used to infer orthogroups with OrthoFinder version 2.3.11 (Emms and Kelly 2015, 2019). We then subsampled these orthogroups to develop two data sets. To determine differences in purifying selection and GC content, we generated a data set composed of putative single-copy orthogroups for *V. appalachiana*, *V. lineata*, and a closely related outgroup, *A. raddianum*. Note that these orthogroups are putatively single copy; since we used transcriptome assemblies, we are unlikely to capture the entirety of gene space for these species, although both assemblies had relatively high BUSCO (benchmarking universal single-copy ortholog) scores. We used the total data set (all 13 species) for gene family evolution and WGD analyses. We extracted the corresponding CDS for each set of orthologs using the python script *extract_cds.py* (Pelosi et al. 2022) and aligned nucleotide sequences using the codon-aware aligner MACSE version 2.04 (Ranwez et al. 2011).

dN/dS Estimation

Pairwise estimates of the ratio of divergence at nonsynonymous and synonymous sites (*dN/dS*, a standard measure for assessing selection pressure on coding regions) were calculated using codeml in PAML version 4.9h (Yang 2007) using the *A. raddianum*, *V. appalachiana*, *V. lineata* putative single-copy ortholog set, with *A. raddianum* as the outgroup (with setting runmode -2). *Adiantum raddianum* is the closest relative for which there is an available transcriptome assembly from 1KP and was therefore used as the outgroup. Orthologs under purifying selection ($dN/dS < 1$) were retained for downstream analyses. For each ortholog, we calculated the difference in *dN/dS* between the *V. appalachiana–A. raddianum dN/dS* value and the *V. lineata–A. raddianum dN/dS* value. We then developed a list of candidate outlier loci by selecting those differences of *dN/dS* values that were above or below 1.5 times the interquartile range (IQR). We performed a Gene Ontology (GO) enrichment test for positive and negative outliers against the background of all genes under purifying selection with the R package topGO version 2.38.1 (Alexa and Rahnenfuhrer 2020) and corrected for multiple comparisons using the Benjamini and Hochberg (1995) false discovery rate (FDR) method. We repeated the *dN/dS* estimation with a concatenated matrix of all loci as above and compared the number of nonsynonymous and synonymous mutations across the matrix with Fisher's exact test (Fisher 1934).

TE Loads

RepeatMasker version 4.0.5 (Smit et al. 2015) was used to generate estimates of repeat content in the transcriptome assemblies of *V. appalachiana* and *V. lineata* using a Viridiplantae repeat set. We also used a read-based approach with dnaPipeTE version 1.4c (Gouber et al. 2015) to account for differences in library size. Since dnaPipeTE uses single-end reads, we used only the forward-read pairs from each library for this approach. CDS for the Gag-Pol and Env proteins, which are specific to retrotransposons, were extracted from the *Arabidopsis thaliana* TAIR10 and *Zea mays* genomes to generate a protein database. We used Blastx with an e-value cutoff of 1×10^{-5} to determine whether Gag-Pol and Env genes were transcribed in the transcriptomes of *V. appalachiana* and *V. lineata*.

GC Content

We calculated the total and third codon position observed GC content (GC3) of putative single-copy ortholog alignments between the *Vittaria* and *Adiantum* ortholog data set using the custom python script *GC_content.py*. Given that base composition could be heterogeneous and evolve slowly (Moores and Holmes 2000) and the loss of sexual reproduction likely occurred recently (Farrar 1990), we also implemented an approach to calculate equilibrium GC content. Alignments of the third codon position were extracted using the custom python script *extract_codons.py* (Pelosi et al. 2022). Total (GC*) and

third codon position (GC3*) equilibrium GC content were calculated using BppML in the Bio++ Suite version 2.3.1 (Dutheil and Boussau 2008) using the method in Galtier and Gouy (1998). Briefly, the nonhomogeneous model proposed by Galtier and Gouy (1998) is a modification of the T92 model (Tamura 1992) with a fixed κ (transitions/transversions ratio) and one θ (GC*) per branch in the tree. We used the *Vittaria* + *Adiantum* single-copy ortholog set to estimate GC* and GC3* of the total and third codon position with a gamma distribution with four rate classes + invariant sites and extracted these values for *V. appalachiana* and *V. lineata* branches. A locus-by-locus approach was used for loci with sufficient branch lengths (loci with inaccurate θ estimates were removed from downstream analyses; loci with $\theta < 0.1$ or $\theta > 0.9$ were discarded). A concatenated alignment of all loci was used to generate a tree with IQTREE2 version 2.1 (Minh et al. 2020) partitioned by locus to get initial branch lengths for the maximum likelihood (ML) search in BppML. Paired Wilcoxon signed-rank tests were used to determine whether differences in GC, GC3, GC*, and GC3* were significantly lower in *V. appalachiana* than in *V. lineata*.

Gene Family Evolution and Identification of Meiotic Genes

Gene trees for single copy orthologs from the full ortholog set were generated using IQTREE2 version 2.1 (Minh et al. 2020) with ModelFinder (Kalyaanamoorthy et al. 2017) and 1000 ultrafast bootstraps (Hoang et al. 2018). A species tree was constructed from these gene trees under the multispecies coalescent using ASTRAL-III version 5.6.2 (Zhang et al. 2018). We inferred expansions, contractions, gains, and losses of gene families under the Wagner parsimony method with a gain penalty of 1.2 (Amborella Genome Project 2013; F.-W. Li et al. 2018), implemented in Count version 9.1106 (Csurös 2010). We then performed GO enrichment analyses as above on each of the gene family sets (expansions, contractions, gains) using the entire *V. appalachiana* transcriptome as the background. For losses, we used a background GO list composed of the *V. appalachiana* transcriptome and *V. lineata* contigs that correspond to the gene families inferred to have been lost in *V. appalachiana* relative to the most recent common ancestor (MRCA) of *V. appalachiana* and *V. lineata*. We corrected for multiple comparisons using the Benjamini and Hochberg (1995) FDR method. All R analyses were performed in R version 3.6.1 (R Core Team 2019).

Genes known to be involved in meiosis and recombination were identified from Schurko and Logsdon (2008) and Osman et al. (2011; app. S1). Peptide sequences for these genes were downloaded from the *A. thaliana* TAIR10 genome assembly on Gramene version 64 (http://www.gramene.org). We then used Blastx with an e-value cutoff of 1×10^{-5} to identify any homologs of these genes that were present in the *V. appalachiana* and *V. lineata* transcriptome assemblies. Blast hits were filtered to generate a one-to-one list of the best Blast hit (based on alignment length, e-value, and bit score) to a transcript.

Divergence Time Estimation and WGD Inference

The program wgd (Zwaenepoel and Van de Peer 2019) was used to plot the distribution of synonymous substitutions per site (K_S) of paralogs for *V. appalachiana* and *V. lineata*. We fitted mixture models over these distributions using the R package mixtools version 1.2 (Benaglia et al. 2009) and identified putative WGD events from the results of the mixture model analyses. To place WGDs in the evolutionary context of the family, with particular attention to *Vittaria*, we constructed a time-calibrated phylogeny using 364 single-copy loci from all 13 species filtered for relative information content in MARE version 0.1.2-rc (Meyer et al. 2011) and long-branch heterogeneity. Divergence time estimation was performed in MCMCTree in PAML version 4.8a (Yang 2007) using a cauchy distribution around the *Heinrichsia cheilanthoides* fossil (Regalado et al. 2019) with a minimum age of 98.7 Myr and a maximum age of 156.39 Myr. We used a maximum root age of 227 Myr for the divergence between Pteridaceae and the outgroup *Asplenium platyneuron* based on the age from Testo and Sundue (2016).

Next, we performed a multitaxon paleopolyploidy search (MAPS) analysis (Li et al. 2015; Z. Li et al. 2018) to further determine where in the Pteridaceae phylogeny putative WGDs occurred. MAPS requires an ultrametric, ladderized species tree as input; to represent major clades and conform to this ladderized tree format, we sampled the Pteridaceae transcriptomes to include only *A. aleuticum*, *G. arizonica*, *C. acrostichoides*, *P. ensiformis*, *V. appalachiana*, *V. lineata*, and the outgroup *A. platyneuron* and extracted orthogroups for which at least one copy of each of these taxa was present. We inputted these data into MAPS to calculate the empirical distribution of duplicated gene trees along the species tree. Average gene birth (λ) and death (μ) rates were estimated using three subsets of 1000 gene families with WGDgc version 1.3 (Rabier et al. 2014) using the geometric mean of gene family size as the root prior based on the Count output. To create a null distribution, we generated three sets of 1000 simulated trees without a WGD using GuestTreeGen from PhyloGenData (Sjöstrand et al. 2013): 1000 with the maximum likelihood values of λ and μ, 1000 with λ and μ at half the ML values, and 1000 with λ and μ at three times the ML values. We then generated 3000 trees as above with a WGD on the branch leading to *V. appalachiana* and *V. lineata* to create a positive distribution with a retention rate of 20%. We compared the MAPS analysis using the empirical data against these simulations using Fisher's exact tests. Given how close the K_S peaks were in the plots of paralogs from *V. appalachiana* and *V. lineata* (see "Results"), we further explored the placement of duplicates in orthogroups for which there were exactly two copies of *V. appalachiana* and *V. lineata* and one copy of an outgroup (either *A. aleuticum* or *A. platyneuron*). We generated gene trees from these orthogroups as above and examined their topology. If duplicates were from a single shared event, we would expect them to fall in one clade, but if they were from two independent events, we would expect *V. appalachiana* duplicates to form a clade and *V. lineata* duplicates to form a separate clade.

Results

Efficacy of Purifying Selection

There were a total of 4982 putative single-copy orthologs shared between *Adiantum raddianum*, *Vittaria appalachiana*, and *Vittaria lineata* for which we obtained *dN*/*dS* values, 4937 of which were under purifying selection (*dN*/*dS* < 1; 99.1%

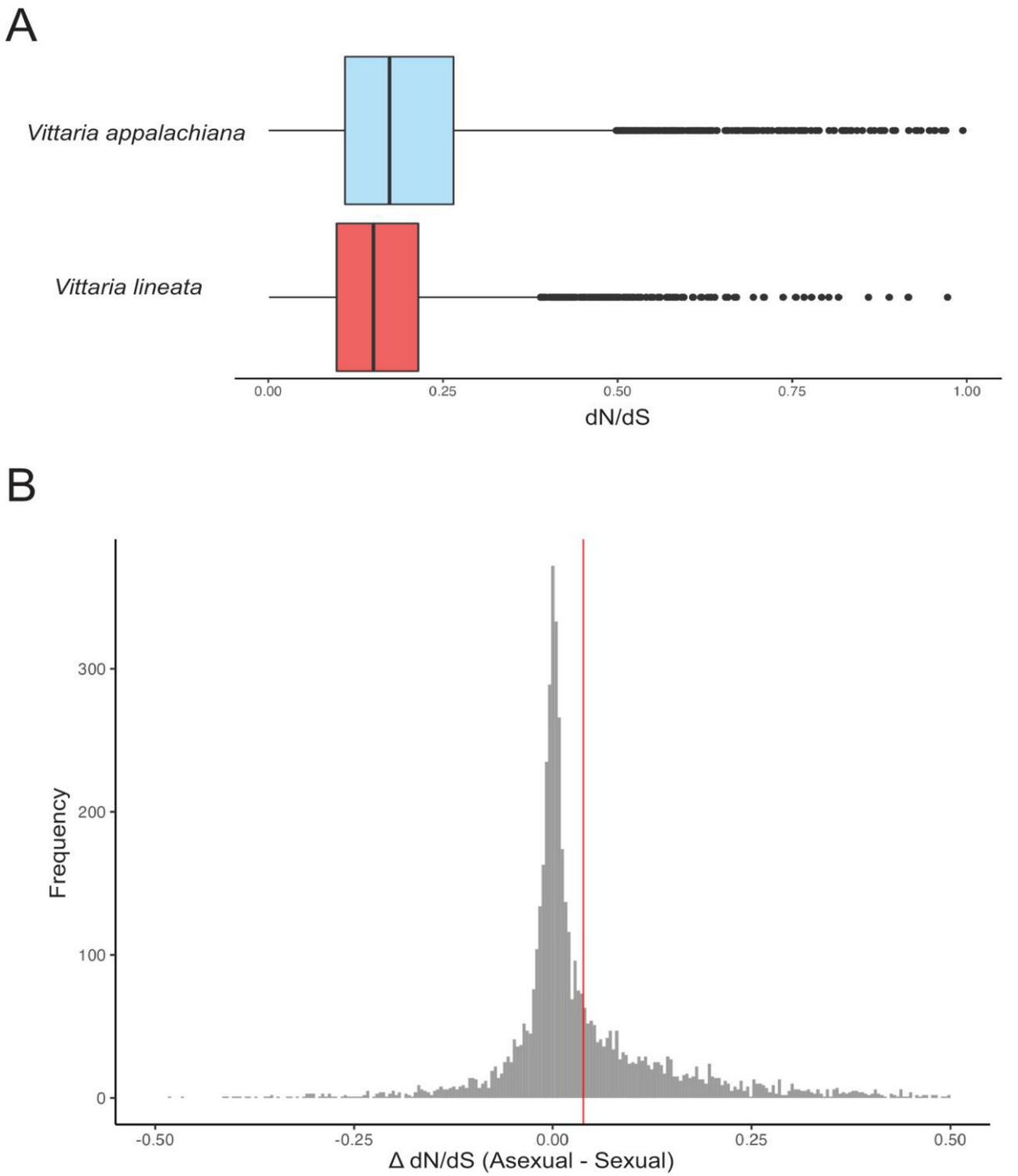


Fig. 2 *A*, Boxplots of *dN*/*dS* ratios for *Vittaria appalachiana* and *Vittaria lineata*. *B*, Differences in *dN*/*dS* ratios (*V. appalachiana*–*V. lineata*) for 4937 single-copy orthologs under purifying selection for *V. appalachiana* and *V. lineata*. Red line denotes the mean difference (mean = 0.038).

of loci). Ratios of *dN*/*dS* for orthologs under purifying selection were significantly greater in *V. appalachiana* compared to *V. lineata* (mean difference = 0.038, Wilcoxon signed-rank test, $P < 2.2 \times 10^{-6}$; fig. 2). There was a total of 527 orthologs with *dN*/*dS* ratios that were above 1.5 × IQR and 182 that were below 1.5 × IQR. There were no significantly enriched terms in either outlier set after correcting for multiple comparisons; however, several interesting terms were significantly enriched in genes under less effective purifying selection (positive outliers) prior to correction, including G2/M transition of the mitotic cell cycle (GO:0000086), cell cycle G2/M phase transition (GO:0046113), and CAF-1 complex (GO:0033186;

Fisher's exact test, $P < 0.01$; see "Discussion"; app. S2). A concatenated approach did not yield significant differences in the number of nonsynonymous and synonymous mutations between *V. appalachiana* and *V. lineata* (Fisher's exact test, $P = 0.2459$), although the *dN*/*dS* ratio for *V. appalachiana* (0.169) was greater than for *V. lineata* (0.164; table 1).

TE Composition and Activity

Based on the RepeatMasker assembly-based approach, a total of 655 Kb (1.22%) of the *V. appalachiana* and 394 Kb (1.13%) of the *V. lineata* transcriptome assemblies were masked as repeats (fig. 3*A*; table 1). Regarding TEs, 177 Kb (0.33%) and 83.8 Kb (0.24%) were retroelements and 46.2 Kb (0.09%) and 32.7 Kb (0.09%) were DNA transposons in the *V. appalachiana* and *V. lineata* transcriptomes, respectively (fig. 3*B*, 3*C*). Repeat content was greater in *V. appalachiana* for all categories, except for the percent of the transcriptome occupied by DNA transposons (fig. 3*C*). Both *V. appalachiana* and *V. lineata* transcriptomes contained the DNA transposons hobo-Activator, Tourist/Harbinger, Mirage, and IS630-Tc1-mariner superfamilies. Families of retrotransposons present in both transcriptomes were short interspersed nuclear elements (SINEs), R2/R4/NeSL, L1/CIN4, Ty1/Copia, and Gypsy/DIRS1, while a single, partial Penelope element was present in the *V. appalachiana* transcriptome but not the *V. lineata* assembly. We found similar results with the raw read-based approach with dnaPipeTE, with 2.01% and 0.44% of the *V. appalachiana* and *V. lineata* libraries being repetitive, respectively (fig. 3*D*, 3*E*; table 1). Approximately 0.59% and 0.21% of the *V. appalachiana* and *V. lineata* libraries were classified as TEs (table 1), respectively. Classification of repeats below the level of DNA and retrotransposon was unsuccessful with dnaPipeTE. There were 219 unique contigs with Blastx hits in the *V. appalachiana* transcriptome (0.205% of expressed contigs) to the Gag-Pol and Env database compared to 95 in the *V. lineata* transcriptome (0.145% of expressed contigs; table 1).

GC Content

There were 5225 loci for which we calculated observed GC and GC3 content. The total GC and GC3 (fig. 4) contents of these ortholog sets were both less in *V. appalachiana* than in *V. lineata* (total mean difference in GC = 0.113%, Wilcoxon signed-rank test, $P = 4.409 \times 10^{-9}$; GC3 mean difference = 0.219%, Wilcoxon signed-rank test, $P < 8.598 \times 10^{-7}$; table 1). Calculations of total GC* and GC3* were made for 3465 and 2298 loci, respectively. Loci with insufficient branch lengths for these estimations were removed, resulting in different numbers of loci analyzed. Both GC* content and GC3* content were also lower in *V. appalachiana* than in *V. lineata* (total mean difference in $GC^* = 0.465\%$, Wilcoxon signed-rank test, $P = 0.043$; GC3* mean difference = 1.163%, Wilcoxon signed-rank test, $P < 4.73 \times 10^{-3}$; fig. 4; table 1).

Gene Family Evolution and the Meiotic Tool Kit

There were a total of 14,700 gene families identified in the data set containing transcriptomes from across Pteridaceae and *Asplenium platyneuron*. A total of 1115 expansions, 671 contractions, 1619 gains, and 1115 losses of gene families were inferred along the terminal branch of *V. appalachiana* (fig. 5). After correcting for multiple comparisons, 11 GO terms were enriched in expanded gene families (including 8 biological process [BP], 0 cellular component [CC], and 3 molecular function [MF] terms), 5 terms enriched in contracted gene families (5 BP, 0 CC, 0 MF), 384 terms enriched in gained gene families (226 BP, 91 CC, 67 MF), and 26 terms enriched in lost gene families (7 BP, 13 CC, 6 MF; Fisher's exact test with FDR

Table 1

Summary Statistics for Several Measures of Asexual Genome Evolution for *Vittaria appalachiana* and *Vittaria lineata*

	Selective pressure			
Species	*N*(loci)	*S*(loci)	*dN*/*dS*(loci)	*dN*/*dS*(cat)
V. appalachiana	*615.9* (257.8–974.0)	*227.4* (96.1–358.7)	*.206* (.064–.349)	*.169*
V. lineata	*619.1* (259.5–978.7)	*224.2* (94.7–353.6)	*.168* (.063–.273)	*.167*
	TE abundance			
	% repetitive	% TE	No. Gag-Pol and Env	
V. appalachiana	1.22/2.01	.42/.59	219	
V. lineata	1.13/.44	.33/.21	95	
	GC content			
	Total GC	GC3	Total GC*	GC3*
V. appalachiana	*46.4* (42.9–49.8)	*43.8* (37.2–50.4)	*51.3* (33.3–69.3)	*48.7* (28.7–68.7)
V. lineata	*46.5* (43.3–49.7)	*44.0* (37.7–50.3)	*51.8* (34.9–68.7)	*49.7* (31.1–68.6)

Note. Measures of selective pressure for individual loci are given as averages of *N*(loci), the number of nonsynonymous mutations for each locus; *S*(loci), the number of synonymous mutations for each locus; and *dN*/*dS*(loci), the rate of nonsynonymous to synonymous mutations for each locus. Measures are given as italicized averages followed by ±1 SD. *dN*/*dS*(cat) values are estimates of the rate of nonsynonymous to synonymous mutations for the concatenated matrix. Repetitive elements are given as percent of the transcriptome assembly/percent of raw reads for the total repetitive element content and transposable elements (TEs). The number of Gag-Pol and Env elements with significant Blast hits is also presented. Measures for GC content are given as italicized averages followed by ±1 SD.

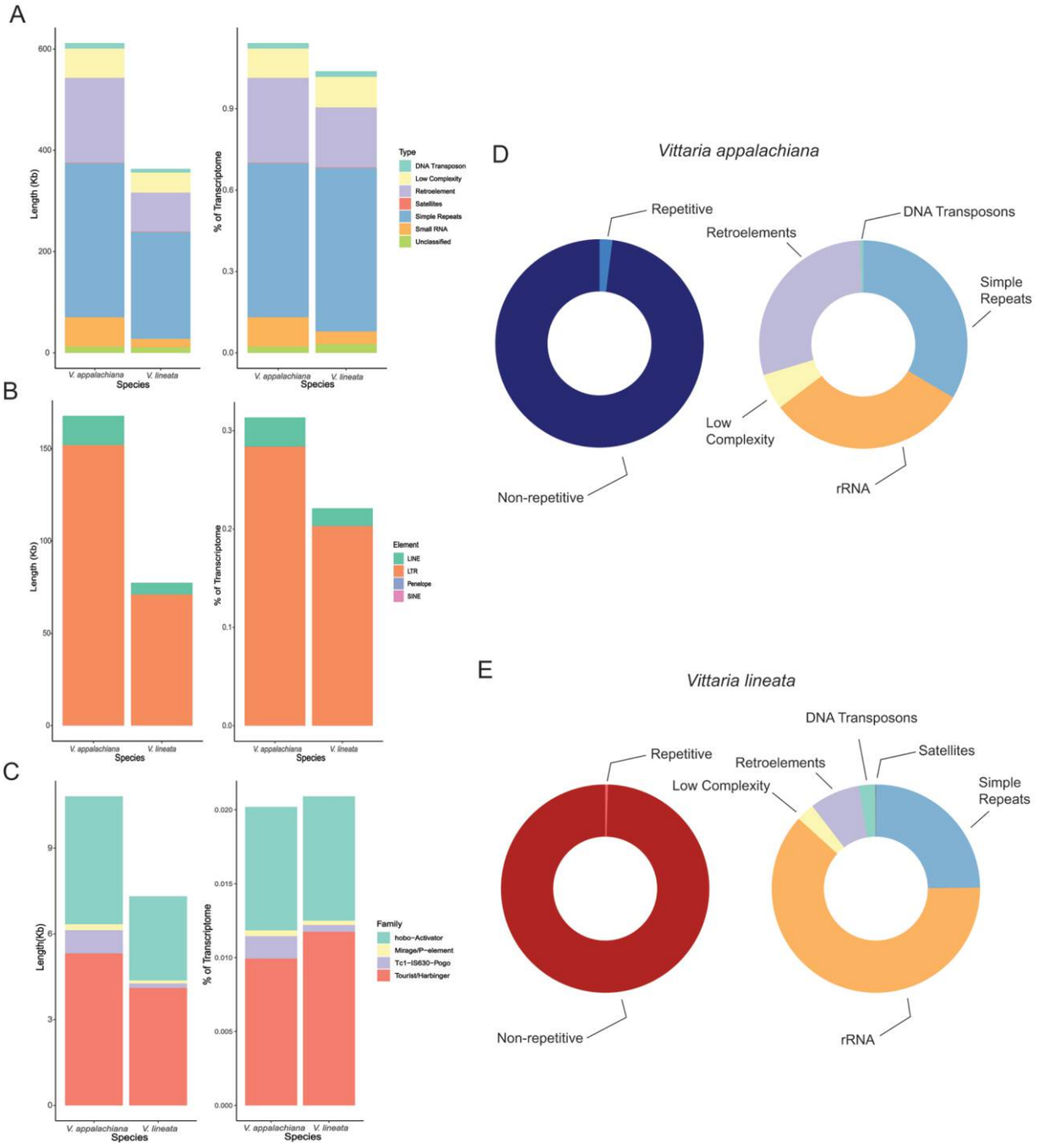


Fig. 3 Length of repeats and percent of the transcriptome masked for repeats in *Vittaria appalachiana* and *Vittaria lineata* for all repeat types in the Viridiplantae RepeatMasker database (*A*), retrotransposons (*B*), and DNA transposons (*C*). Proportion of reads classified as repetitive (left donut plot) and proportion of repetitive reads identified as different repeat types (right donut plot) for *V. appalachiana* (*D*) and *V. lineata* (*E*).

correction, $P < 0.01$; app. S3). Of the 31 genes in the *Arabidopsis* meiotic tool kit, we identified Blast hits to 21 in the *V. appalachiana* transcriptome and 17 in the *V. lineata* transcriptome (app. S1). All 17 hits in the *V. lineata* transcriptome were also present in the *V. appalachiana* assembly, but 4 meiotic genes (MSH5, MLH3, NBS1, XRCC3) were present in *V. appalachiana* but not *V. lineata* (app. S1). Ten meiotic genes were without hits in either transcriptome (MSH4, PRD1, PRD2/MPS1, PRD3, PTD, SHOC1, SPO11-2, ZIP4, ZYP1a, ZYP1b).

Paleopolyploidy Events

We inferred two putative WGDs in both *V. appalachiana* (K_S peaks at 0.51, 2.40) and *V. lineata* (K_S peaks at 0.58,

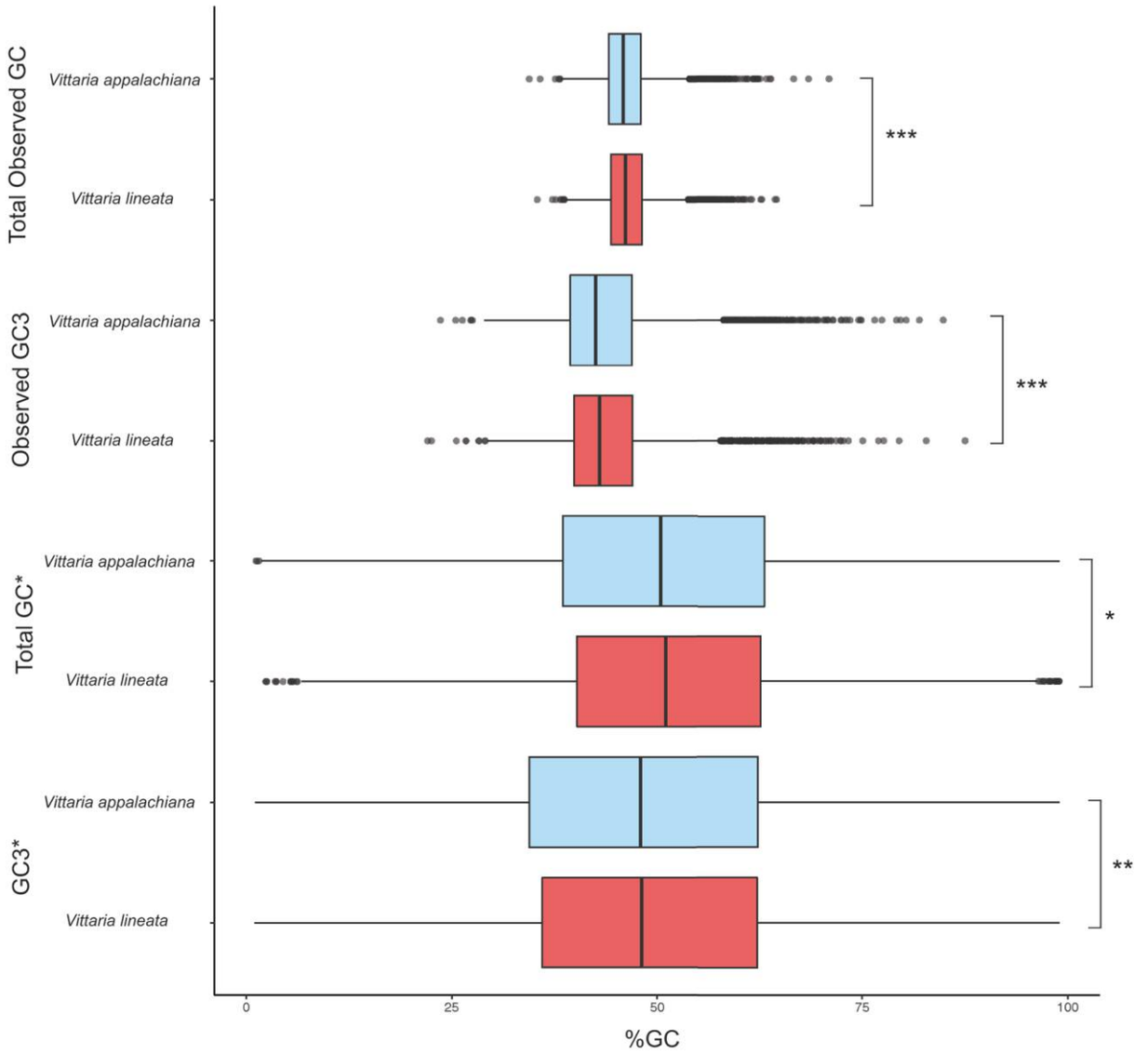


Fig. 4 Differences in observed and equilibrium total and third codon position GC content of single-copy orthologs for *Vittaria appalachiana* and *Vittaria lineata*. Significance: $*P < 0.05$; $**P < 0.01$; $***P < 0.001$.

2.41; fig. 6*A*, 6*B*). Based on the placement of the duplicate OGs (see "Methods") and the MAPS analysis, we putatively placed these WGDs on the branch leading to Pteridaceae and before the MRCA of *V. appalachiana* and *V. lineata* (fig. 6*A*). There was a significantly greater proportion of duplicated subtrees compared to the null distribution at the MRCA of *V. appalachiana* and *V. lineata* (Fisher's exact test, $P = 4.170 \times 10^{-13}$) but not at the base of Pteridaceae (Fisher's exact test, $P = 0.807$; app. S4). Furthermore, the proportion of duplicated gene trees at the MRCA of *V. appalachiana* and *V. lineata* was not significantly less than expected based on our positive MAPS simulations (Fisher's exact test, $P = 0.937$; app. S4). Our additional tests to determine whether the WGDs were separate or shared between *V. appalachiana* and *V. lineata* identified 56 and 53 gene trees that had *V. appalachiana* duplicates sister to each other (fig. 6*C*), suggesting separate WGDs, with *Asplenium* and *Adiantum* as the outgroups for the analyses, respectively. In contrast, there were 457 and 396 gene trees that supported a topology where *V. appalachiana* and *V. lineata* duplicates were sister, suggesting a shared WGD (fig. 6*C*), with *Asplenium* and *Adiantum* as the outgroups for the analyses, respectively.

Discussion

Vittaria appalachiana is a gametophyte-only fern that is well suited to address fundamental questions regarding long-term asexual (i.e., vegetative, clonal) reproduction in the gametophytic life stage in plants. Using publicly available data, we found support for theoretical genomic consequences of asexual

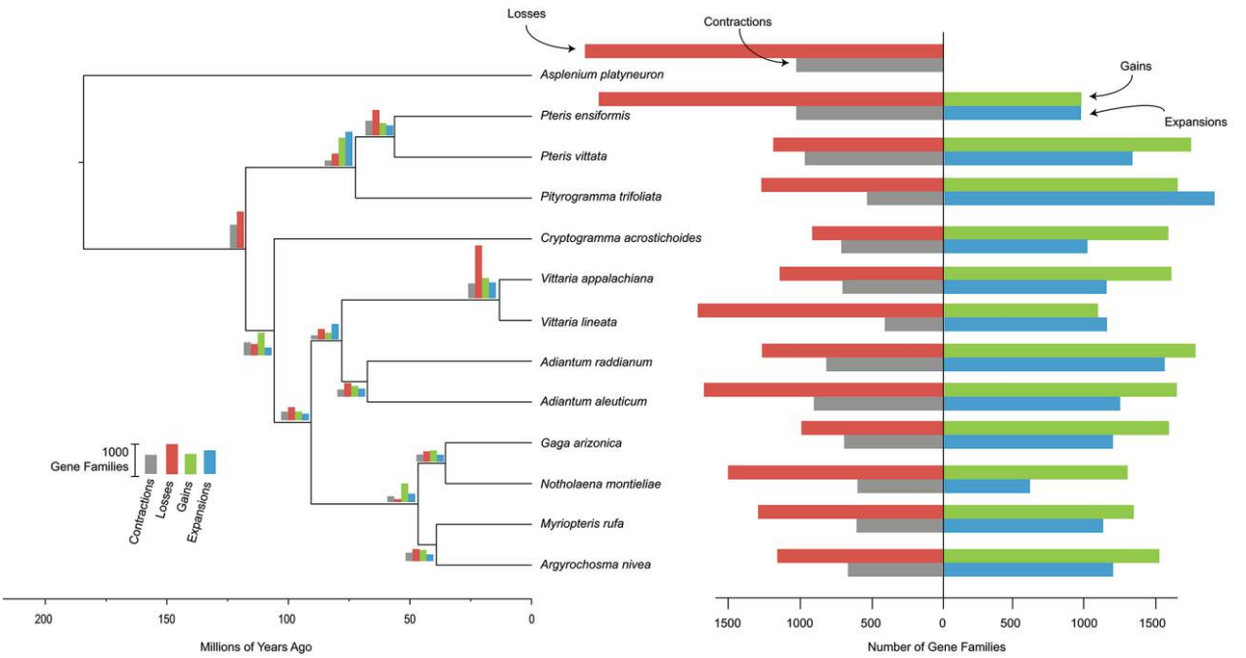


Fig. 5 Gene family evolution across Pteridaceae. Expansions (blue), gains (green), losses (red), and contractions (gray) of gene families are shown for nodes on the tree (*left*), with tip values represented in the bar plot (*right*). The full time-calibrated phylogeny can be found in figure S1.

reproduction in *V. appalachiana*. Genomic alterations due to asexual reproduction may take millions of years to accumulate and are more likely to be apparent in ancient asexuals (Neiman et al. 2009; Jaron et al. 2021) compared to lineages that have recently lost sexual reproduction. The timing of the putative loss of *V. appalachiana* sporophytes is uncertain, with hypotheses ranging from the Pliocene (ca. 2.6 Ma) to the Illinoian glaciations (ca. 191,000–130,000 years ago; Farrar 1990), and it is possible that the heightened rate of molecular evolution previously estimated in vittarioid ferns (Rothfels and Schuettpelz 2014; Grusz et al. 2016) has sped up this process in this species.

Before discussing these results, we must acknowledge a few caveats of the data that constitute limitations or confounding factors the reader should be aware of while interpreting our results. All transcriptomes are snapshots of gene expression at one time in one tissue and do not necessarily capture the entirety of gene space. While genomes would be ideal for a study like this, the haploid genome size of *V. appalachiana* is 33.99 pg (Pelosi et al., forthcoming), and, at current costs, sequencing and assembling a genome of this size is prohibitively expensive. Transcriptomes are substantially smaller and therefore more feasible for studies of this kind. The data used in this study came from the publicly available assemblies produced by 1KP (2019), and at the time of writing there were no available transcriptomic data available for the sister species (*Vittaria graminifolia*) of *V. appalachiana*. Rather, we used the assembly of *Vittaria lineata* as the sexual representative to which we compared genomic features of *V. appalachiana*. It is possible that differences observed between the two species occurred prior to the evolution of asexuality and polyploidy in *V. appalachiana*. Furthermore, while the transcriptome of *V. appalachiana* comes from gametophyte tissue, the transcriptome of *V. lineata* is derived from sporophytic tissue. Ploidy differences between gametophytic (diploid *V. appalachiana*) and sporophytic (tetraploid *V. lineata*) tissues may therefore be important factors to consider; interestingly, however, sporophyte and gametophyte tissues in ferns have been found to exhibit little differentiation in gene expression (Sigel et al. 2018; Marchant et al. 2022), with most genes being expressed at similar levels in both tissues and very few genes being expressed in one tissue only. Both *V. appalachiana* and *V. lineata* are of a polyploid origin, and the multiple genomes present in tetraploid, sporophytic *V. lineata* could influence selective pressure on duplicate genes and partly mask the effect of asexual reproduction on genome evolution. For these reasons, there is limited power to assign a causal relationship between differences in these genomic features and asexual reproduction. Despite the limitations of these data, they represent the only publicly available genomic or transcriptomic data set of an asexual/sexual species pair with which to explore the potential effects of asexual reproduction on the genome structure of a fern. The generalizability of these results should be considered carefully until further investigations generate more directly comparable data sets.

Decreased Efficacy of Purifying Selection

Using a locus-by-locus approach, we determined that genes in *V. appalachiana* were on average under less effective purifying selection than in *V. lineata*. While the concatenated approach found no significant differences in the number of nonsynonymous and synonymous mutations between the two taxa (although the *dN/dS* ratio was greater for *V. appalachiana* than

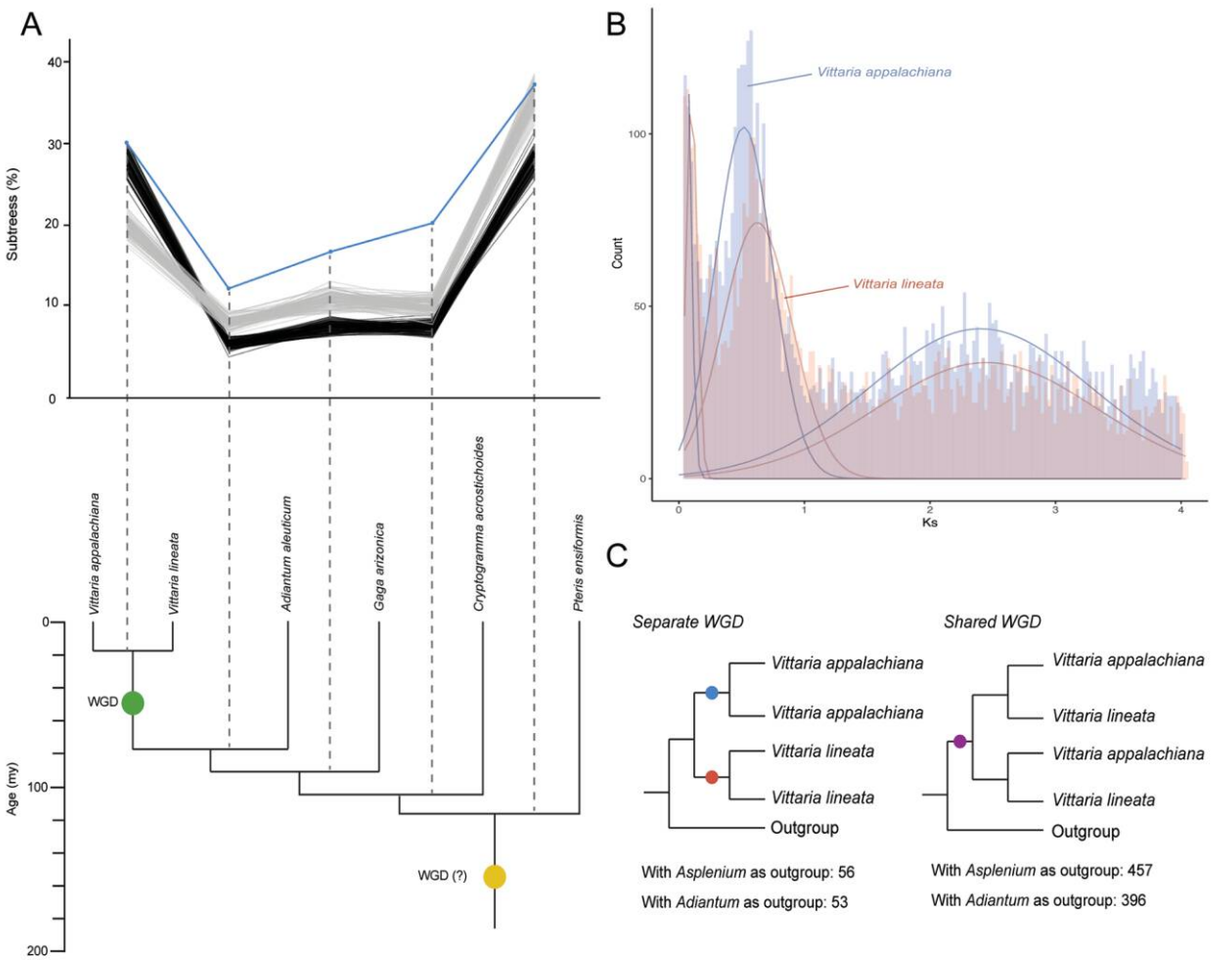


Fig. 6 *A*, Multitaxon paleopolyploidy search analysis of whole-genome duplications (WGDs) in Pteridaceae (blue line), with corresponding ultrametric phylogeny for taxa included in the analysis. The gray lines correspond to 100 null simulations based on birth-death processes; the black lines correspond to 100 positive simulations with a WGD placed on the branch leading to *Vittaria appalachiana* and *Vittaria lineata*. The percentage of subtrees with duplications is given on the *y*-axis with dashed lines connected to the corresponding nodes. The two inferred WGDs are denoted with circles; the green circle corresponds to the shared *Vittaria* WGD, and the yellow circle corresponds to a potential duplication shared in Pteridaceae. Circles are placed arbitrarily along these branches and do not correspond to the age of the WGD. *B*, Paralog-age distributions (K_S plots) for *V. appalachiana* (blue) and *V. lineata* (red), with fitted mixture models. *C*, Gene duplicate topologies for separate WGDs in *V. appalachiana* and *V. lineata* (*left*) and a single shared WGD before the most recent common ancestor of *V. appalachiana* and *V. lineata* (*right*). Hypothesized WGDs are denoted with circles (blue, red, purple). The corresponding number of gene trees supporting each topology is given for each topology with *Asplenium platyneuron* or *Adiantum aleuticum* as the outgroup. The full time-calibrated phylogeny can be found in figure S1.

for *V. lineata*; table 1), this method can blur conflicting signals from single loci. Genes with putative functions related to the progression of the cell cycle (app. S2), including G2/M transition of the mitotic cell cycle (GO:0000086), regulation of the G2/M transition (GO:0010389), and CAF-1 complex (GO:0033186), were found to be enriched in loci under less effective purifying selection. We did not find support for the hypothesis that genes involved in sexual reproduction should decay more rapidly than the rest of the genome as they have been released from selective constraints (Normark 2003). The accumulation of deleterious mutations in genes is a specific demonstration of Muller's ratchet; mutations will accumulate over time without recombination to shed deleterious variants. Given our results, it is possible that *V. appalachiana* may eventually succumb to the effects of deleterious mutants and become extinct, as is expected to be the case in most asexually reproducing taxa. However, relaxed selection is a common phenomenon in asexually reproducing organisms, including ancient asexually reproducing animals that have persisted for millions of years (reviewed by Hartfield 2016). Not all ancient asexuals, however, exhibit this pattern of relaxed purifying selection (Welch and Meselson 2001; Brandt et al. 2017).

Changes in selective pressures have been observed in other asexually reproducing organisms (Johnson and Howard 2007; Neiman et al. 2010; Hollister et al. 2015; Bast et al. 2018), although not all have shown decreases in the efficacy of purifying selection (e.g., Brandt et al. 2017). For example, Simion et al.

(2021) sequenced the genome of the asexually reproducing bdelloid rotifer *Adineta vaga* and found footprints of homologous recombination despite a lack of cytological evidence for meiosis. Using recombination, coupled with horizontal gene transfer and large population sizes, *A. vaga* is able to purge deleterious alleles while maintaining heterozygosity and has persisted for over 60 Myr without sexual reproduction. Could the same or similar processes be at play in *V. appalachiana*? Could infrequent mitotic recombination between chromosomes retained from WGDs in a *V. appalachiana* provide a way to purge deleterious alleles? Alternatively, deleterious alleles may be masked by the presence of several genomes, and infrequent or absent recombination could maintain fixed heterozygosity, although isozyme patterns do not suggest the presence of multiple expressed alleles (Farrar 1990). Like other homosporous ferns with high chromosome numbers, *V. appalachiana* has diploid gene expression (Farrar, 1985), as suggested by Haufler and Soltis (1986). Further genomic and cytological study is needed to determine whether these or other processes occur in *V. appalachiana* and to what extent they are possibly driving the continued persistence of the species in the face of the genomic consequences of asexual reproduction.

TE Loads

TEs make up a nontrivial proportion of the transcriptomes of both *V. appalachiana* and *V. lineata* (fig. 3; table 1). This is not surprising, as repetitive elements make up the majority of plant genomes (Wendel et al. 2016), with ferns generally having greater repeat content compared to seed plants (Wolf et al. 2015). The increased proportion and number of repetitive elements in the *V. appalachiana* transcriptome compared to *V. lineata* (fig. 3; table 1) may reflect that the efficacy of selection to remove TEs in this asexually reproducing species is less than in the sexually reproducing *V. lineata*. Similar TE families were present in the gametophyte transcriptome assembly of *Pteridium aquilinum* (Der et al. 2011), although both *V. appalachiana* and *V. lineata* had SINEs identified and did not have rolling circle helitrons, MuDR-IS905, or PiggyBac DNA transposon families. The expression of retrotransposon-specific genes in both transcriptomes points to active TEs in these genomes, with perhaps more activity in *V. appalachiana* compared to *V. lineata* given the greater number of Blast hits (table 1). Alternatively, TE expansions may reflect increases due to polyploidy (McClintock 1984; reviewed by Wendel et al. 2016). Our results conflict with recent findings that parthenogenic animals tend to have lower TE loads compared to sexually reproducing animals (Bast et al. 2019; Jaron et al. 2021) or do not have different loads (Bast et al. 2016). The transcriptomic data used here may be biased in that they are likely under greater purifying selection than the rest of the genome, which is noncoding. The insertion of a TE into or near a CDS could disrupt the normal function of that gene (McClintock 1950), whereas an insertion into a nonfunctional sequence elsewhere in the genome could have a lesser impact (Sigman and Slotkin 2016). While the implications from these findings are enticing, biases in RNA sequencing must be considered. Although RNA libraries are enriched for polyA sequences, residual genomic DNA may also be sequenced (e.g., Li et al. 2022). The data collected here may represent these (non)random genomic DNAs. Therefore, further work should focus on characterizing the repeat landscape throughout the genome to truly assess differences in the abundance of TEs and their impact. While developing a whole-genome assembly for these two species may currently be out of reach, genome-skimming data may be a viable alternative for further exploring these repeat families in a fuller genomic context (e.g., Wolf et al. 2015; Ruiz-Ruano et al. 2021).

Decreased GC Content

Gene conversion during meiosis tends to favor increases in GC content (Marais 2003; Duret and Galtier 2009). Therefore, in asexually reproducing organisms that lack meiotic recombination, GC content should theoretically decrease when compared to a closely related sexually reproducing relative. We found that *V. appalachiana*, which reproduces entirely asexually, has a significantly lower transcriptomic observed and equilibrium GC content compared to its sexually reproducing relative *V. lineata* (fig. 4; table 1). The difference between the GC content of *V. appalachiana* and *V. lineata* is quite small; although these differences are statistically significant, whether this difference is biologically relevant requires further examination. In general, recombination is not uniform throughout genomes and instead occurs with greater frequency in "hot spots" (Thuriaux 1977; Brown and Sundaresan 1991; reviewed by Choi and Henderson 2015). The small difference seen in our data may reflect that we have not captured these hot spots and therefore regions under gBGC if it is occurring, although recombination hot spots in *Arabidopsis thaliana*, *Triticum aestivum*, and *Zea mays* have been found in genic regions (Choi and Henderson 2015). Furthermore, differences in GC content may be impacted by codon usage bias; given the redundancy of the genetic code, translational machinery in taxa may differ and "prefer" certain codons over others. Additional study is required to investigate the role of codon usage bias in driving GC content in ferns. To date, ferns have not been included in analyses of biased gene conversion (but see Li et al. 2016 for evidence for gBGC in the inverted repeat of the plastome in ferns), and recent work with plants overall shows strong evidence for gBGC (Pessia et al. 2012; Rodgers-Melnick et al. 2016; Clément et al. 2017; Hämälä and Tiffin 2020). With new resources for ferns becoming available in the genomic era (e.g., F.-W. Li et al. 2018; Qi et al. 2018; Shen et al. 2018; One Thousand Plant Transcriptomes Initiative 2019; Huang et al. 2022; Marchant et al. 2022), investigating whether this phenomenon occurs in these plants will be readily addressable. In particular, the model fern *Ceratopteris richardii* may be an ideal candidate for testing gBGC, as there is a wealth of literature on mutant and recombinant cultures (reviewed in Hickok et al. 1995), a genetic map (Nakazato et al. 2006), and a nuclear genome (Marchant et al. 2022) available for this species.

Life History Changes Could Drive Gene Family Evolution

Our analyses suggest that patterns of gene family evolution could be related to changes in reproductive mode. We identified 384 GO terms enriched in gained gene families in *V. appalachiana* relative to the MRCA of *V. appalachiana* and *V. lineata*, with functions largely related to cell cohesion (GO:0015629,

GO:0008092, GO:0042995, GO:0030031) and cell morphology (GO:0098743, GO:0031589, GO:0007155). We also identified several GO terms that were enriched in expanded gene families associated with Mo-molybdopterins (GO:0006777, GO:0019720, GO:0043545), a set of cofactors that are related to the modulation of cold and osmotic stress response gene expression in *Arabidopsis* (Bittner et al. 2001). Interestingly, the gametophytes of several tropical *Vittaria* species can tolerate cooler temperatures at latitudes 34°–42°N (Farrar 1978; Chambers and Emery 2016), whereas their sporophytes cannot (Farrar 1978). Whether these expanded gene families are related to the presumed cold tolerance of *V. appalachiana* remains to be thoroughly tested and warrants further attention. These gains of genes related to morphological and metabolic components relative to a sexually reproducing ancestor with a sporophyte are also accompanied by putative losses in gene families related to sporophyte-specific morphology (GO:0071994, GO:0009505, GO:0071554) and DNA repair machinery (GO:0006281, GO:0006974). Additional study is needed to assess whether the gene families identified here are truly lost from the genome, pseudogenized, or simply not represented in the transcriptome assembly and whether function of these genes is conserved between ferns and angiosperms.

Retention of the Meiotic Tool Kit

Homologs of most genes in the meiotic tool kit were expressed in both the *V. appalachiana* and *V. lineata* transcriptomes (app. S1). Meiotic genes that were identified in *V. lineata* were also found in *V. appalachiana*, although 10 of the 31 standard genes occurred in neither transcriptome. The *Polypodium amorphum* transcriptome (Sigel et al. 2018), which was derived from six libraries of both gametophytic and sporophytic tissue, contains 29 of the 31 meiosis-related genes. Two genes, Hop1 and MSH6, exhibited differential expression in the sporophyte and gametophyte, respectively. Deeper sequencing of the *Vittaria* transcriptomes may reveal the presence of the 10 currently absent genes. Missing genes from the *V. appalachiana* and *V. lineata* transcriptomes are also involved in the formation of double-stranded breaks (PRD1, PRD2/MPS1, PRD3/ATPRD3, SPO11-2) and the primary crossover pathway (class I; MSH4, PTD, SHOC1, ZYP1a, ZYP1b, ZIP4; Osman et al. 2011). Although meiosis-related genes should be under less effective purifying selection and therefore decay over time (Schurko and Logsdon 2008), we were able to identify most genes involved with meiotic recombination. Hanson et al. (2013) also identified retained genes in the meiotic tool kit in monogonont rotifers and found that several genes were duplicated in these asexually reproducing species and showed differential expression between sexually and asexually reproducing relatives. Using putatively meiosis-related genes to discover "sex scandals" (instances where sexual reproduction is ongoing but not observed in theoretically obligate asexually reproducing organisms) in asexually reproducing organisms has been criticized (Maciver 2019). Rather than being an indication of cryptic sexual reproduction, the detection of genes in the meiotic tool kit in asexually reproducing organisms can occur because they have important functions in other pathways. For example, MRE11 is involved in repairing double-stranded DNA breaks in both meiosis and mitosis (Tsubouchi and Ogawa, 1998) and RAD51 functions in telomere maintenance (Cho et al. 2014). Interestingly, although gene families contracted or lost in *V. appalachiana* were enriched for DNA repair overall (see above), meiotic DNA repair genes specifically are still present; this could indicate that while other DNA repair genes are dispensable, this core set that is involved in meiosis is not. While meiotic genes are conserved across eukaryotes (de Massy 2013), very few functional molecular biology experiments related to fern reproduction are available, although recent transcriptome analyses found some of the genes above are expressed in meiotic fern tissues (Lu et al. 2018). The functions of the putative homologs of meiotic genes in ferns require further study to determine whether they are retained across evolutionary time.

A Complicated Origin and the Role of Polyploidy

Our analyses suggest that one WGD occurred before the MRCA of *V. appalachiana* and *V. lineata*, in addition to an older event possibly shared by Pteridaceae (fig. 6). Although the distribution of duplicated gene trees for the older event is not significantly greater than the null distribution, this event could correspond to the WGD shared by Polypodiales (Pelosi et al. 2022). The event shared by both *Vittaria* species agrees with the inferences by Pelosi et al. (2022), who found a shared WGD at the base of the vittarioid ferns. This duplication would account for the transition from $n = 30$ to $n = 60$, but both *V. appalachiana* and *V. lineata* have $n = 120$ (Gastony 1977), suggesting that additional WGD(s) may have occurred within *Vittaria*. The available chromosome counts for this genus are sparse (Pinson and Schuettpelz 2016), and, coupled with the taxonomic uncertainty in *Vittaria*, make it difficult to determine whether a second WGD is present and whether it is shared by *V. appalachiana* and *V. lineata* or whether there were independent polyploidization events. There are several hypotheses about the origin of *V. appalachiana* revolving around polyploidy. Using a phylogenetic approach, Pinson and Schuettpelz (2016) proposed that *V. appalachiana* is an autopolyploid of *V. graminifolia*, while Farrar (1990) used population genetic data to suggest an allopolyploid origin with *V. graminifolia* as one of the putative parents. The genetic variation observed in some populations (Farrar 1985, 1990) could support multiple allopolyploid origins (D. R. Farrar, personal communication). Despite the findings here and the phylogenetic framework developed by Pinson and Schuettpelz (2016), additional chromosome counts, ploidy estimates, and genomic data are still clearly needed to resolve the origin of the Appalachian gametophyte.

Conclusions

While there is an inherent difficulty in assessing genome-wide evolutionary trends from transcriptomic data, especially in nonmodel organisms, we found putative support for predicted genomic consequences of asexual reproduction (e.g., decreased efficacy of purifying selection, altered TE load, decreased GC content) in *Vittaria appalachiana*. We further identified two putative WGDs, shared by Pteridaceae and the vittarioid ferns, which may have contributed to *V. appalachiana*'s persistence over evolutionary time without sexual reproduction. These data give the first look into the genomic consequences of asexual

reproduction and evolution of a gametophyte-only fern and suggest several avenues for future research.

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Data Availability

All code is available at https://github.com/jessiepelosi/vittaria_trans.

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